

**Nijmegen School of Management
Department of Economics and Business Economics
Bachelor's Thesis in Economics (MAN-BIM304A)**

How does the relationship between AI adoption and economic growth vary across levels of economic development?

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Nijmegen, 22 May 2026

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Abstract

This paper's main focus is on the relationship between AI adoption and economic growth and how this relationship changes across different economic developmental levels. The motivation behind this research is to address the growing concern that AI adoption might result in increased global inequality, because developed countries may benefit more from it than less developed nations. Furthermore, the theoretical framework of this text incorporates endogenous growth theory, diffusion of innovations theory, the view that AI might be a general-purpose technology and much more to create its initial hypothesis. This hypothesis states that the effects of AI adoption on economic growth should be conditional on economic development and that more advanced economies should have a more positive effect than less advanced economies.

In both regression models, the interaction term between AI adoption and the Human Development Index variables were significantly positive, meaning that the effect of AI adoption becomes more positive as countries have higher HDI scores, in other words when they are more developed. These findings mostly correlate with the initial hypothesis and the existing literature in large. In conclusion, the findings suggest that AI-driven economic growth is distributed unequally between more and less developed economies and that only countries with the adequate structural and institutional environments can benefit substantially from this AI-driven technological change.

1 Introduction

Artificial Intelligence (AI) is predicted to transform labour markets and entire economic sectors around the world (Cazzanige et al., 2024). Bold predictions such as these are based on credible research done by major institutions, such as the International Monetary Fund (IMF) and credited academics. Moreover, the number of people using generative AI tools, such as ChatGPT has been growing and it already reached 3 billion visits per month in 2024 (Wang & He, 2024) and according to ChatGPT's own latest estimate, this number reached 5.8 billion monthly visits in September of 2025 with approximately 800 million weekly active users (Singh, 2025).

This technology has the potential to lead to productivity growth in many industries, further resulting in higher economic growth (Damioli et al., 2024). Empirical data supports the claim that AI technologies has been significantly correlated with firm-level productivity in the period between 2002 and 2018 in Taiwan (Yang, 2022). Another example of productivity growth by AI can be seen in the financial industry. AI tools can fundamentally change the financial industry with massive advances in firm-level productivity (Golic, 2019). In other words, a significant amount of literature has been published on AI's effects on productivity-levels in an economy. According to economic theory, these productivity gains should result in increased economic growth in the sectors affected by this technology (Babina et al., 2024).

Evidence for the potential of AI-driven economic growth can be found in many economic papers. However, the distribution of the benefits of this growth is not a main area of concern in much of the literature. Furthermore, the distributional effects of any disruptive technology such as this on economic growth is expected to be unequal, as it was before with other technologies in history (Batabyal et al., 2025). Close to 40 percent of all global employment could be exposed to AI in the future (Cazzanige et al., 2024). Out of this employment loss it is not clear which regions in the world can lose the most jobs and which regions will gain the most benefits of the transformation.

There are different challenges that countries face due to their differences in how their economies have developed. While economically advanced nations are more exposed to initial economic shocks caused by artificial intelligence, they are the ones that are expected to receive the most out of the benefits of this added economic growth produced by it as well (Cazzanige et

al, 2024). Poorer economies might not be as vulnerable in the short-term, but the long-term effects are very much uncertain (Cazzanige et al., 2024).

Despite the enormous potential of AI-driven economic growth, recent economic research suggests that the benefits will not be distributed evenly between economies, workers and sectors overall, this could result in the widening of global wealth inequality across nations (Minniti, 2025). Dynamic general equilibrium analysis reveals that AI-driven economic growth could almost entirely be enjoyed by the most economically advanced nations, especially in the short-term (Liu, 2025). These findings are alarming from social, economic and ethical perspectives. In addition, understanding what inequality risks AI could bring in the future is crucial for predicting the true extent of the effects of AI adoption on the global economy.

Therefore, the main aim of this article is to do research about what the distribution of AI-driven economic growth around the globe will likely to be with a specific focus on economic development as a measure to differentiate between economic regions. Especially thinking about the already deep regional and economic inequalities around the globe, it is important to ask what happens to countries who are not as developed as the economically advanced nations, who are leading this technological transition.

Since the research on AI is still relatively scarce due to the situation and the facts around it are constantly changing, this article is meant to contribute to the existing literature with a special focus on the differences in economic outcomes based on a country's development. Key points that differentiate this research and therefore that extend the existing body of literature are a cross-country perspective, a focus on linking AI's economic effects to economic development and a perspective that analyses AI's distributional effects between countries.

The paper's objective is to create a regression based on publicly available data, for example on AI adoption, GDP growth and the Human Development index. This regression measures AI-driven economic growth with economic development as one of the broad independent variables. The structure of the paper is as follows, section 2 contains the literature review, section 3 is the methodology, section 4 will contain the results of the regression model and finally section 5 contains the conclusions about the results and an overall review of the article's findings.

2 Literature Review

2.1 Defining Artificial Intelligence

There are several definitions of Artificial intelligence (AI) present in the literature, one of these is the definition invented by Alan Turing, where artificial intelligence was described as “the science and engineering of making intelligent machines” (Turing, 1950). Furthermore, this article will use the most common definition for AI derived by Russel and Norvig, where AI is defined as something that “enables the machine to exhibit human intelligence, including the ability to perceive, reason, learn, and interact with beings” (Collins et al., 2021). Moreover, since this paper examines the effects of the most commonly used and adopted artificial intelligence systems, the term AI will be used to refer to systems, such as large language models (LLMs), for example ChatGPT and BERT as well as other AI systems that are the most popular computer systems in the market at the specific period of analysis (Bender & Koller, 2020). Technological advances in machine learning and natural language processing (NLP) have made AI more integrated into economic processes, which raises important questions about the economic consequences of this change (Bender & Koller, 2020).

2.2 AI and Economic growth

Although many major institutions focus on this topic, such as the IMF, research around the effects of AI on economic growth is still very limited, since the wide-scale adoption of AI into economic processes around the world is a relatively recent phenomenon. However, there are various studies researching the subject nevertheless, for example some articles focus on the possibility that the introduction of AI can lead to a technological paradigm shift, which can be comparable to inventions such as the internet (Damioli et al., 2024). Much of the current discussion about AI revolves around precisely this theory, that these systems will lead to productivity growth in many sectors of the economy, such as finance or even in agriculture, which will potentially result in increased economic growth (Bjola, 2022; Golic, 2019). This theory fits into the idea of innovative diffusion, where technologies get adopted that exhibit relative advantage gains, compatibility with existing economic processes and observability of results (Rogers, 1983). All of these factors influence rate of adoption, but the speed of this AI adoption remains uncertain,

considering the nature of technological progress in general and it is influenced by certain economic indicators such as general infrastructure or institutional capacity (Bjola, 2022).

One particular sector for example that AI could transform is the financial sector. Due to the financial industry's tendency to be digitally advanced in analytics processing, the sector is one of the earliest adopters of AI technologies, the expected benefits are projected to be enormous, for example the ability to provide better and faster services to clients at lower costs, so essentially productivity gains caused by AI (Golic, 2019). Firms that adopt AI tools are more likely to develop new products and have higher productivity growth than firms that did not adopt these technologies (Babina et al., 2024). Moreover, firm-level analysis further provides evidence for AI driven economic growth (Yang, 2022).

2.3 AI adoption and Inequality

Despite these promising findings, most researchers agree that the distribution of the economic benefits of AI is highly unequal across countries with different economic development levels. AI usage is mostly concentrated in highly educated demographic groups, which favours developed countries (Yan & He, 2024). Cross-country analysis reveals that wealthier and more urbanized regions exhibit higher levels of AI adoption (Goel & Nelson, 2025). Middle- and high-income countries account for the vast majority of the global usage of this technology (Khan et al., 2024). These findings suggest that there could be structural limits for low-income countries in their ability to economically gain from the AI transformation.

Furthermore, this uneven adoption of AI technologies raises important issues around the potential for increased global inequality that could arise from this AI transformation. Countries with higher levels of human capital are projected to benefit more from using AI tools (Batabyal et al., 2025). Similarly, there is a symbiotic relationship between human capital and AI, which means that productivity growth by AI is largely dependent on the worker's ability to use AI tools (Gomes, 2025). Moreover, labour markets could radically transform according to this logic, reducing demand for so called middle-productivity jobs (Cazzaniga et al., 2024). This transformation could create more economic inequality within and between countries. Several studies reach similar conclusions about the potential for increased wealth inequality. AI has the potential for productivity growth in different sectors, but it could also replace certain processes entirely (Fleire, 2025). In addition, wealth distribution inside and between countries could change, because AI could alter

capital accumulation dynamics inside firms and sectors (Liu & Liang, 2025). To summarize, AI-driven economic growth is very likely dependent on the general economic development of a given region.

2.4 Endogenous Growth theory

There are many economic theories about how technological innovation can lead to sustained long-term economic growth. General neoclassical economic theory treats technological progress as an exogenous factor that develops outside the economic system. This can be called exogenous growth theory. In contrast, Romer's (1990) endogenous growth theory focuses on the idea that technological advancements are key drivers themselves of long-run economic growth, in other words he treats technology as an endogenous variable when it comes to modelling long-term economic outcomes. In his theory, technological innovations, such as AI tools function as non-rival economic goods, which means that many agents in an economy can use them as a resource without actually losing their initial value, in other words without depleting their fundamental value. Furthermore, in this framework innovation and knowledge accumulation can function as key drivers of sustained long-term GDP-growth.

In addition, artificial intelligence as a technological tool can be seen as a non-rival good according to this theory. Specifically, AI has the capacity to increase economic productivity by increasing the efficiency of information processing and knowledge production. AI systems can automate many cognitive tasks and in turn they can accelerate economic innovation. Many papers focus on this topic by connecting endogenous growth theory to AI as a non-rival technological good. Aghion et al. (2019) argues that AI has the ability to automate certain production and innovation processes acting as a major driver of economic growth. Moreover, Cockburn et al. (2018) states similar conclusions, it hypothesises that AI could revolutionize technological progress itself and in turn could speed up technological innovation in the future. Romer's (1990) perspective on long-term economic growth and the role of scientific progress is relevant to the research question of the paper, since it implies that a country's ability to benefit from new innovations can depend on endogenous factors, such as education levels, institutional quality and infrastructure. This essentially means that economic outcomes could differ largely between different levels of economic development across markets.

2.5 AI as a General-purpose technology

Economists classify different kinds of tools, innovations and resources in the economy based on their functions and potential use cases. One of these classifications is general-purpose technologies, which are defined as technologies that can be used in many different sectors of the economy and therefore, they can fundamentally change production systems as a whole. Specific innovations that are examples for this idea are electricity, the steam engine and the internet. Moreover, some recent scientific articles link AI to this concept and they argue that it could be a kind of general-purpose technology. They state that AI has the potential to completely disrupt production systems in the future as the reason for this classification (Brynjolfsson et al., 2021).

General-purpose technologies can only generate productivity growth, once the required investments in the economy are made, for example in the case of electricity, places that use electricity today have built the adequate infrastructure to be able to use this resource. In the case of AI, something similar could happen, where only the regions that have developed the required institutional frameworks and infrastructures could benefit from AI and the regions that are unable to do this could stagnate. Therefore, the level of AI adoption can be influenced by how developed a country is and this will have an effect on economic growth prospects. In other words, this theory further substantiates the claim harnessing the potential benefits of AI could be conditional on economic development (Brynjolfsson et al., 2021).

2.6 Technological Diffusion and Institutional quality

Rogers' (1983) diffusion of innovations theory further supports these findings. It states that technologies get adopted faster as they exhibit clear productivity gains and compatibility with already existing infrastructures and networks. This process depends heavily on economic, social and institutional conditions in a particular country or region. In AI's case, adoption clearly depends on the level of digital infrastructure. Therefore, countries with low digital literacy, weak digital infrastructure and general low levels of human capital may find it difficult to adopt AI technologies at even a comparable rate as countries that do not face these problems do.

Furthermore, an important issue that is relevant for the adoption of AI is institutional quality and responses from policy makers. Countries that have strong institutions, functioning legal and political systems can enact strong protections and regulations when it comes to AI technologies.

On the other hand, countries with weak institutions will not be able to create similar innovation-friendly conditions that could result in increased AI-driven economic growth. Consequently, the role of institutions and policy can also influence the economic outcomes of adopting AI technologies in an economy.

2.7 Summary and Hypothesis

To summarize, theories around economic growth and technological innovations, such as endogenous growth theory, technological diffusion theory and the theory about general-purpose technologies all suggest that AI adoption rates are linked to economic development and the economic effects of this technological adoption could be different across countries with differing levels of economic development.

This article will examine this relationship between AI and economic growth, especially focusing on the role of economic development and who and which regions can benefit more from this technological advancement. Based on the review of the available literature on this subject, the paper's theoretical framework and initial hypothesis for its analysis is that the effect of adopting AI systems into employment-related processes on economic growth is dependent on a region's economic development. In other words, the hypothesis is that there is a positive relationship between AI driven economic growth and the level of economic development.

3 Methodology

3.1 Research Design

This research paper is a quantitative study, and it uses a particular regression model, based on the theoretical framework and hypothesis. The hypothesis of this paper predicts that there is a positive relationship between AI driven economic growth and economic development. The timeframe of the analysis is given based on partially data availability and historical dates that are set based on the initial research framework. The period when all data was collected is between 2004 and 2025, including data from both 2004 and 2025. The reason for this choice is that google trends started collecting data in 2004 and therefore this is the earliest date that the dataset could still have data for. While early AI systems have been used before this date, the period after 2000 and then after 2010 (Collins et al., 2021) is considered the timeframe that the current AI transformation as we know it started and is still evolving in. In addition, the analysis uses data from as many nations as possible, which are ended up being the nations that had investment data.

The paper uses panel data that contains both cross-sectional and time-series data, this allows for comparing the different outcomes between countries and throughout different years. This is very important, because the economic effects of AI adoption should show major variation between countries based on time and economic development. The regression uses fixed effects. Country fixed effects are useful to control for characteristics that are country-specific, in other words effects that remain constant over-time for a particular country, this could be for example cultural or geographical effects. Moreover, time fixed effects are applied to control for specific events that are irregular, this could be for example a global recession. Using a quantitative regression model is justified for this paper, because it incorporates specific macroeconomic statistics, such as investment, human development index and GDP growth. Using panel data models when modelling economic growth is considered a regular practice in the literature (Fernández-Val & Weidner, 2018).

3.2 Variables and Data sources

The table below depicts all variables that are used in this regression and their respective sources (Table 1). All datasets are referenced, but there are small adjustments that were made in the data, so there could be more valid observations. For GDP growth, there is no data for 2025, so the value for 2025 is the same as it was in 2024, this strategy is repeated for the HDI index and the investment control variable. The regression uses the investment control variable and time and country fixed effects to have more accurate and reliable results.

The dependent variable in the regression is GDP per capita growth, which is one of the most common economic indicators for this kind of analysis. This captures economic performance better than total GDP growth would, because it takes in account population size and can better measure productivity changes in a country.

The main independent variable in this analysis is the AI google trends data. Specifically, the google trends variable uses the google search scores to account for variations of AI searches that occurs over-time. The google search scores are from the Google trends website, and the data was collected by looking at individual countries for individual years and it collects scores for every full year (2004-2025), this was done via Python, cause manually it would be inconvenient and time consuming (Google Trends, 2026). The variable is standardized for the regression, because this way the analysis can present more accurate results. The analysis uses this variable to measure the economic and social interest for AI technologies over time. Its value represents how aware and ready a population is for adopting AI technologies in economic processes.

Another independent variable is the human development index, which captures the economic development of countries. In addition, the "artificial intelligence" Google search scores and the HDI variable will form an interaction variable. Lastly, the investment variable is used as a control variable, specifically this measures the gross annual capital formation as a percentage of GDP. This variable is included, because capital accumulation is an important predictor of economic growth, therefore it serves as an adequate control variable.

Table 1. Variables Table

Variables	Full Name	Source
$GDPpcGrowth_{\{i,t\}}$	GDP per Capita Growth	GDP per Capita Growth data by the World Bank
$AI_trends_std_{\{i,t\}}$	AI google trends data	Google trends data about the term “artificial intelligence”
$HDI_{\{i,t\}}$	Human Development Index	HDI by UNDP
$Investment_X_{\{i,t\}}$	Investment Control Variable (basic macroeconomic indicator)	World Bank Group Gross Capital formation (% of GDP)
μ_i	Country Fixed Effects	Included in the model.
λ_t	Time Fixed Effects	Included in the model.
$\varepsilon_{\{i,t\}}$	Error term	No source needed

3.3 Regression Equation

Furthermore, the regression equation of for the variables and the analysis is presented in this section (1 Regression Equation). The interaction term between AI trends and HDI is the main focus of the paper, it is meant to test the effects of AI adoption on economic growth, and it varies across economic development levels.

(1) Regression Equation

$$GDPpcGrowth_{\{i,t\}} = \beta_1 AI_trends_std_{\{i,t\}} + \beta_2 HDI_{\{i,t\}} + \beta_3 (AI_trends_std_{\{i,t\}} \times HDI_{\{i,t\}}) + Investment_X_{\{i,t\}} + \mu_i + \lambda_t + \varepsilon_{\{i,t\}}$$

Notes: Dependent Variable = GPT per Capita growth

β_1 = Measures the effect of AI trends variable on the dependent variable

β_2 = Measures the effect of HDI index on the dependent variable

β_3 = Measures the effect of the interactive variable between HDI and AI trends on dependent variable.

All references and coding are included in the references, specific coding can be accessed via the GitHub link at the very end of the paper.

3.4 Descriptive Statistics and Correlation Matrix

For all the variables used in this regression, there is a descriptive statistics table and a correlation matrix included for the dependent and independent variables. The software used to process data and create the regression is python. Mult correlation problems cannot be detected based on the correlation matrix table. For the descriptive statistics table, for the variables GDP growth and investment, there are a few outliers that make these variables volatile. However, this is expected since these statistics can be very different across countries and time periods.

Table 2. Descriptive Statistics Table

	observations	mean	std	min	25%	50%	75%	max
gdp_pc_growth	2616	2,0835	5,1069	-49,1279	0,2326	2,1794	4,2510	91,7814
ai_trends_std	2616	-0,1013	0,8624	-1,8209	-0,6189	-0,3247	0,1189	4,5826
hdi	2616	0,7322	0,1589	0,2790	0,6090	0,7630	0,8620	0,9720
investment	2616	22,9347	6,8414	1,2252	18,8317	22,2781	25,8064	70,1086

Table 3. Correlation Matrix Table

	gdp_pc_growth	ai_trends_std	hdi	investment
gdp_pc_growth	1	-0,0498	-0,0289	0,1905
ai_trends_std	-0,0498	1	0,0367	-0,0440
hdi	-0,0289	0,0367	1	0,0423
Investment	0,1905	-0,0440	0,0423	1

3.5 Limitations of the Methodology

An earlier model used AI preparedness scores that were calculated by the IMF. However, there was a multicollinearity problem with the regression between the AI Readiness Index (ARI) and the Human Development Index (HDI), these variables both capture some indicators of general economic development and therefore show a significant level of correlation between the values. Because of this reason, the actual equation ended up not using these scores. This made the

1 correlation much more effective and Google trends alone can still function as a key indicator of
2 AI adoption rates and implementing AI technologies.

3 Furthermore, an important limitation is the problem with heteroskedasticity. The dataset
4 includes very different countries with different growth volatility and economic structures, this
5 makes it unlikely that the variance of the error term can remain constant across observations.
6 Robust standard errors are used to partially account for this issue.

7 An additional potential problem could be endogeneity. This is because higher economic growth
8 itself might increase AI adoption as well, since wealthier economies have higher capacity for
9 investment. Therefore, it is possible that there is reverse causality between AI adoption and
10 economic growth. This issue is partially addressed by adding country and time fixed effects to the
11 regression.
12

4 Results

This section presents the key empirical findings of this paper. Specifically, it focuses on results about how the adoption of AI influences economic growth. In addition, how this relationship varies across different levels of economic development. Economic development is measured by the Human Development Index (HDI) and economic growth is measured by GDP per capita growth. The analysis includes results from two main regression models. The baseline model includes AI trends, HDI and an interaction term between the two as independent variables. It also includes both time and country fixed effects. The extended model further incorporates investment as a control variable. The main variable of interest is the interaction term between HDI and AI trends variables. This term is meant to test the hypothesis of the paper, which is that the effect of AI adoption on economic growth is conditional on the level of economic development. The analysis first examines the relationship between AI and economic growth, and it also includes the question of whether this relationship is dependent of a country's level of human development.

4.1 Marginal effects of AI adoption

Figure (2) illustrates the marginal effect of AI adoption on expected GDP-per-capita growth and its variation across economic development measured by HDI. The upward sloping line indicates a clear linear relationship. This essentially means that as countries reach higher levels of economic development, the effect of AI adoption on GDP growth increases.

First of all, at low levels of economic development (with HDI around 0.3-0.5) the marginal effect on economic growth is negative. This implies that AI adoption is associated with short-term growth reductions. This could be explained by adjustment costs, insufficient infrastructure, weaker digital capacity or lower levels of human capital. In these countries, AI technologies could still be a topic of interest, but because of structural limitations, they are unable to translate this into actual productivity gains in specific sectors of their economies.

Furthermore, as HDI increases, the marginal effect on economic growth increases and at high levels of economic development it becomes positive. Around an HDI of 0.7 marks the point, where

AI adoption begins to show a positive effect on expected economic growth. This could be an indication that we can adopt our hypothesis since this is exactly the effect that we would expect based on it. Countries with higher HDI levels have better education systems, better infrastructure and more stable institutions, based on literature and our hypothesis, the expectation is that this will result in higher economic growth attributed to AI technologies for these highly developed economies.

These results are consistent with our hypothesis and several economic theories that were explored in the literature review. Firstly, endogenous growth theory states that technological advancements only cause economic growth, if there are sufficient human capital and infrastructure to support it. In addition, technological diffusion theory is also consistent with these results, which suggests that new technological innovations cannot spread evenly throughout countries around the world, instead they can create massive inequalities due to them having very different effects on countries with different institutions, legal systems, economic conditions and general infrastructure. Therefore, the graph that depicts the marginal effect of AI adoption depending on the level of economic development is highly consistent with the literature and theoretical claims made in this paper (Figure 2).

(2) Marginal effect of AI adoption across levels of economic development

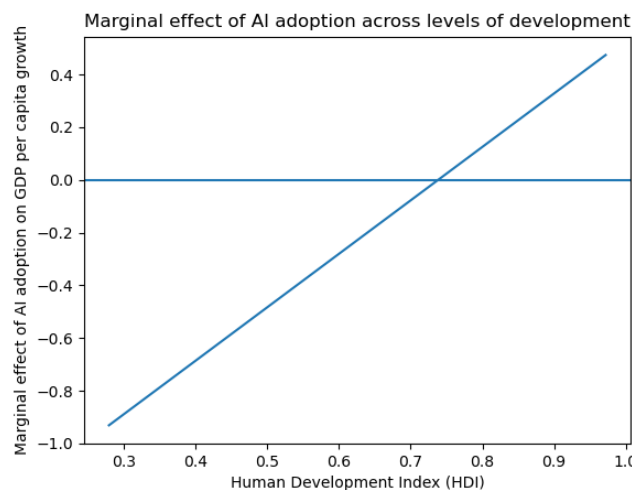
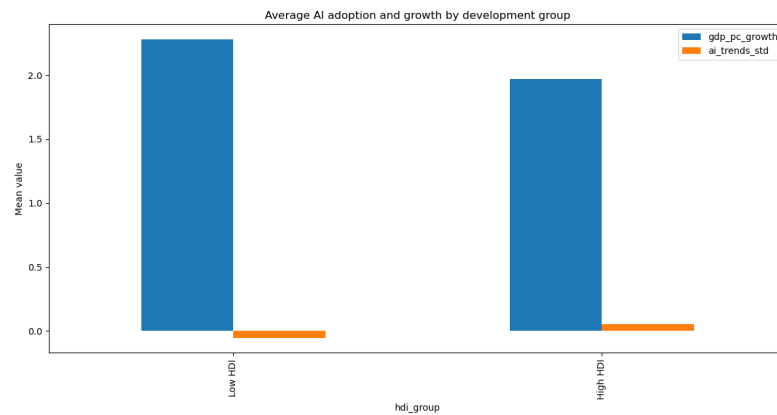


Figure (3) shows the same results in a different way. These results are illustrated by a bar chart, which looks at how AI adoption influences economic growth. It shows that on average while

economic growth is higher for low-income countries (showed by blue bars), the effect of AI adoption is negative for low HDI countries and positive for high HDI countries (indicated by orange bars). It is important to show this distinction, that while low-HDI countries are generally expected to have higher GDP per capita growth than high-HDI nations due to different macroeconomic factors and convergence effects. However, AI still seems to have more favourable effects in high-HDI countries than in low-HDI ones. This is in line with the hypothesis that AI adoption does not necessarily result in the same levels of economic growth attributed to it across economies.

(3) Average GDP growth and AI trends by Development group



4.2 Baseline Model

Column (1) of table 4 reports on the results of the first model. The coefficients on AI adoption, both AI trends (AI_trends_std) and the interaction term between AI and HDI remains significant on the 5% level. The coefficient of AI trends is negative, which suggests on average that AI adoption is associated with lower short-term economic growth. However, this number cannot account for the heterogeneity across countries and different levels of HDI.

This negative coefficient does not necessarily mean that AI adoption has a harmful effect on economic growth. It just means that when HDI is constant at zero, the baseline effect of AI trends variable is negative. Since this condition of HDI being zero is not realistic for most, if not all countries, this coefficient should be considered with the interaction term's coefficient in mind. In other words, the total effect of AI trends variable depends on its coefficient and the interaction

with the HDI coefficient. Moreover, the coefficient for human development (HDI) is positive and highly significant, which is consistent with economic theory. This positive coefficient of HDI supports the idea that HDI is strongly related to economic growth outcomes.

On the other hand, the interaction term between AI adoption and HDI is positive, which indicates that AI adoption has a positive effect on GDP growth when it is conditional on economic development (HDI). As the paper stated before, looking at this coefficient, it is clear that the effect of AI trends becomes positive as HDI increases. This result is the most important finding in this regression analysis, since it is directly supporting the thesis's hypothesis.

To summarize, AI adoption has a negative average effect on economic growth when it is only considered alone, but this effect becomes more positive as we reach higher levels of economic development. This finding suggests that the benefits of adoption AI technologies cannot be distributed equally amongst countries and that they depend on a countries development level.

4.3 Extended Model

Column (2) of table 4 reports on the results of the second model that controls for effects of investments. As investment is introduced as a control variable, the explanatory power of the model increased measured by R squared increasing from 0.306 to 0.328. This mean that the second model can explain slightly better the variation in GDP per capita growth than the first model. The investment control variable that was added in the extended model has a positive and highly significant coefficient. This is consistent with the economic literature that indicates a positive relationship between economic growth and investment levels in a country.

Although the interaction term between HDI and AI adoption loses significance, it still remains significant on the 10 % level. This result reinforces the finding again that the effect of AI adoption is conditional on economic development. The fact that the interaction term stays positive and significant after controlling for investment suggests that the findings cannot be explained by differences in capital formation between countries alone.

The coefficient for AI trends remains negative in the extended model. However, the interaction term remained positive, therefore this again suggests that for higher HDI countries, the positive interaction effect outweighs the negative AI trends effect. Moreover, the coefficient of HDI

decreased and it lost its statistical significance. The reason behind this could be that what HDI originally captured in the baseline model, it is now partially captured by the investment variable. Developed countries have high HDI and also higher levels of investment. Therefore, after the control variable was introduced into the regression, the effect of HDI alone became smaller and not significant.

Overall, this extended model still supports the main hypothesis of the paper. The paper's alternative hypothesis can be accepted based on these results, although the conclusions should still be stated cautiously.

Table 4. Regression Table

	(1) Baseline	(2) + Investment
Ai_trends_std	-1.348** (0.656)	-1.497 (0.927)
Hdi	30.031*** (10.289)	15.631 (9.741)
Ai_x_hdi	1.763** (0.815)	2.028* (1.164)
Investment		0.162*** (0.029)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	3095	2616
R-squared	0,306	0,328
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

4.4 Robustness and Limitations

The results are largely consistent with the hypothesis and the theoretical framework, but there are major limitations that must be stated. The most important issue is about the fact that AI adoption is meant to be measured by Google trend data. This variable mostly captures the search

1 interest and general awareness about AI technologies rather than actual statistics about firms or
2 individuals implementing AI tools. The reason for this is that this kind of data is not available for
3 a thesis paper such as this, this is especially true when we consider that the regression uses data
4 from many different countries. Therefore, the results should only be looked at with this
5 distinction in mind, and it must be acknowledged that this result is about AI-related interest and
6 economic growth, which is not a perfect measure of AI adoption, in reality it can only function as
7 an indicator of it.

8 Secondly, the regression might still suffer from omitted variable bias. There could be various
9 other factors, such as trade openness, institutional quality and readiness, quality of the education
10 system and research and development spending that could affect AI adoption and economic
11 growth. Adding country and time fixed effects and including investment as a control variable helps
12 with this issue, but it cannot fully eliminate it either.

13 Thirdly, the regression looked at a relatively short time period between 2004 and 2025. This
14 might not be enough time to model the long-term economic effects of AI. This is especially true
15 when we consider that generative technologies only became particularly visible recently.
16 Therefore, this result may capture changes in AI interest and possible adoption within a so called
17 early-stage transitional period. This issue is very likely to have an impact on these results, but
18 nevertheless the analysis could only use data from this time period.

19 Considering these limitations, the results should be interpreted with caution. While the key
20 findings show that AI trends, HDI and GDP per capita growth have strong associations in this
21 model, this should not be confused with empirical evidence of a definite causal relationship.

22 **4.5 Summary of the Results**

23 To summarize these results, it can be said that these findings support the general hypothesis of
24 the paper. The baseline model and the extended model reveal similar results. In both models, the
25 interaction term between HDI and AI trends variables were positive and significant, suggesting
26 that the effect of AI adoption on economic growth becomes more and more positive as HDI
27 increases and countries reach higher levels of economic development. Both figure (4) and table 5
28 are meant to illustrate these key findings.

(4) Predicted GDP Growth effect of AI trends by HDI level

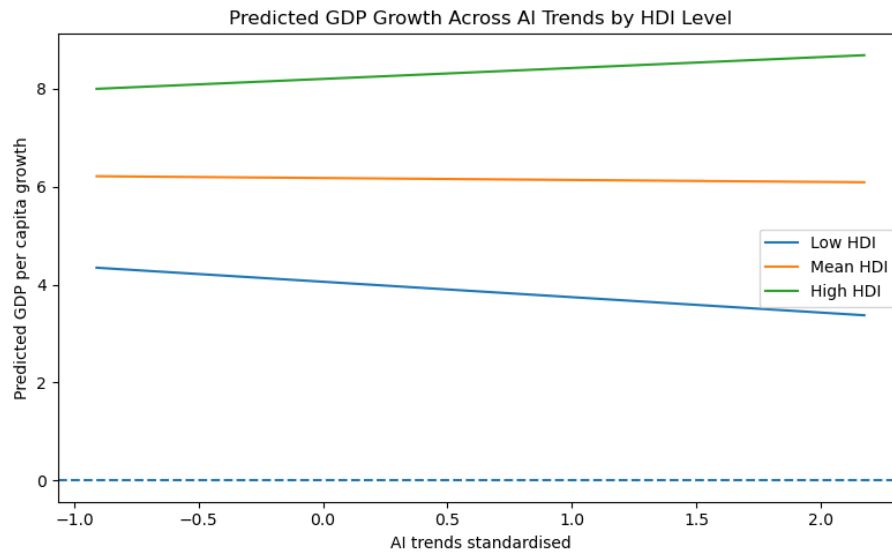


Figure (4) shows the predicted GDP growth effects of AI trends across different HDI levels, the figure differentiates between low, mean and high HDI levels.

Table 5. Marginal effects of AI trends at different HDI levels.

HDI level	HDI value	Marginal effect of AI trends
Low HDI (25th percentile)	0,583	-0,315
Mean HDI	0,719	-0,04
High HDI (75th percentile)	0,848	0,223

Lastly, table 5 shows the same results, but only in a table form with the actual numbers that the model predicts. The low HDI countries have negative (-0,315) marginal effect of AI trends, the mean HDI countries have close to zero (-0,04) and high HDI countries have a positive (0,223) marginal effect.

5 Conclusions

The central focus of this paper was on how the relationship between the adoption of artificial intelligence(AI) technologies and economic growth differs across various economic developmental levels. Despite AI's advantages such as the potential for productivity growth in several sectors of the economy, there is a growing number of papers that research the possible disadvantages that could arise from adopting these technologies. Specifically, the possibility that this technological innovation could further increase global wealth inequality between countries is the motivation for this research to examine the validity of such claims.

Initially, the paper explored many different research papers and economic theories to establish its theoretical background. Specific theories such as endogenous growth theory, technological diffusion theory, the idea that AI could be a general-purpose technology and more were used in this process. From this literature review, came the paper's hypothesis, which states that economic growth that is attributed to adopting AI is conditional on economic development. In other words, as countries become more developed, they should be more able to take advantage of the potential gains from AI innovations.

The empirical analysis mostly supports this hypothesis's validity. In both the baseline and the extended regression models, the results largely remained similar. The interaction term between AI trends and HDI remained positive and significant in the two models. This finding suggests that AI adoption's effect on GDP per capita growth increasingly becomes positive as HDI increases. On the other hand, the coefficient for only AI adoption was negative, but this does not mean that AI adoption could harm economies. Quite the contrary, this just reenforces the idea that AI adoption alone does not necessarily result in higher economic growth, if the country is not developed enough economically. This was further demonstrated by separating the results for low, mean and high HDI countries, while low and mean HDI countries were predicted to have a negative effect of adopting AI, high HDI countries had a positive effect on economic growth.

There are several theories that could prove explanations for this finding. Countries with high HDI scores usually have better functioning institutions, a more educated workforce and a more efficient digital infrastructure, these conditions create more opportunities for them to take advantage of AI's productivity enhancing potential. On the other hand, lower HDI levels usually

mean that these economies are still developing and that even if they adopt AI technologies, partially because of various macroeconomic factors and the lack of functioning institutions they are still unable to fully gain substantial economic benefits from implementing them, especially in the short term. Furthermore, this could also provide an explanation for the negative coefficient of AI trends in both models, since it says that at lower HDI-levels AI adoption could even be harmful for economic growth. This could be because of large adjustment costs without producing any meaningful increases in GDP growth.

Another theory that is consistent with these findings is Romer's endogenous growth model. This model highlights the importance of structural and institutional factors when modelling economic growth. In this view, AI has to be adopted in the right structural and institutional environment to produce significant economic gains. Furthermore, diffusion of innovation theory states that the adoption of highly advanced technologies occurs unevenly across different developmental levels.

The concern that AI could increase global inequality comes up when analyzing these results. The implication is clear, advanced economies can benefit substantially more from adopting these technologies, thereby widening the economic gap between developed and developing nations. This finding contributes further evidence to the growing research around this topic that this concern remains valid and should be looked at seriously. However, the results do not imply that less developed countries can never take advantage of this technological innovation, rather that they will be unable to do so as long as they do not possess the right kind of institutions and structural frameworks to adopt them. This suggests that they might still benefit substantially, but only in the longer term.

In addition, this study has some major limitations, which were discussed in the previous section. There are some ways that this research could be improved and expanded in the future. Most importantly, getting data from actual firms and investments or statistics around AI adoption rates could measure the variable of AI adoption in a more precise way than only looking at google trends data. Furthermore, future research could take advantage of more data on AI adoption that could become available overtime.

1 In conclusion, the key results of this thesis suggest that the effects of AI adoption on economic
2 growth could be strongly conditional on a country's economic development. With these findings,
3 the paper contributes to the existing literature in many ways. Firstly, the analysis has a cross-
4 country perspective, meaning that it has panel data across many countries in the period between
5 2004 and 2025. Secondly, its focus is on the interaction between AI adoption and economic
6 development, while most papers still tend to focus on AI adoption alone. Lastly, the regression
7 provides findings about the possible distributional effects of AI-driven economic growth between
8 nations. This topic remains relatively underexplored in the current literature.

References

- 1
- 2 Acemoglu, D. (2025). The Simple Macroeconomics of AI. *Economic Policy*, (121), 13–58. (2154534).
- 3 *AI Preparedness Index (API)*—*AI Preparedness Index*. (n.d.). Retrieved February 11, 2026, from
- 4 https://www.imf.org/external/datamapper/AI_PI@API
- 5 *AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking*. (n.d.). Retrieved
- 6 April 16, 2026, from <https://www.mdpi.com/2075-4698/15/1/6>
- 7 *Artificial Intelligence and Economic Growth* | NBER. (n.d.). Retrieved May 22, 2026, from
- 8 <https://www.nber.org/papers/w23928>
- 9 *Artificial intelligence*—*Explore*—*Google Trends*. (n.d.). Retrieved February 11, 2026, from
- 10 [https://trends.google.com/trends/explore?date=2004-01-01%202026-01-](https://trends.google.com/trends/explore?date=2004-01-01%202026-01-01&q=Artificial%20intelligence&hl=en-us)
- 11 [01&q=Artificial%20intelligence&hl=en-us](https://trends.google.com/trends/explore?date=2004-01-01%202026-01-01&q=Artificial%20intelligence&hl=en-us)
- 12 Babina, T., Fedyk, A., He, A., & Hodson, J. (2024). Artificial intelligence, firm growth, and product
- 13 innovation. *Journal of Financial Economics*, 151, 103745.
- 14 <https://doi.org/10.1016/j.jfineco.2023.103745>
- 15 Batabyal, A. A., Kourtit, K., & Nijkamp, P. (2025). Artificial intelligence based technologies and
- 16 economic growth in a creative region. *Economics of Innovation and New Technology*, 34(5), 782–
- 17 796. <https://doi.org/10.1080/10438599.2024.2374273>
- 18 Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On Meaning, Form, and Understanding in
- 19 the Age of Data. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th*
- 20 *Annual Meeting of the Association for Computational Linguistics* (pp. 5185–5198). Association for
- 21 Computational Linguistics. <https://doi.org/10.18653/v1/2020.acl-main.463>

- 1 Bjola, C. (2022). AI for Development: Implications for Theory and Practice. *Oxford Development*
2 *Studies*, 50(1), 78–90. (1963166). <https://doi.org/10.1080/13600818.2021.1960960>
- 3 Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E. J., & Tavares, M. M.
4 (2024). Gen-AI: Artificial Intelligence and the Future of Work. *Staff Discussion Notes*, 2024(001).
5 <https://doi.org/10.5089/9798400262548.006.A001>
- 6 Collins, C., Dennehy, D., Conboy, K., & Mikalef, P. (2021). Artificial intelligence in information systems
7 research: A systematic literature review and research agenda. *International Journal of*
8 *Information Management*, 60, 102383. <https://doi.org/10.1016/j.ijinfomgt.2021.102383>
- 9 Damoli, G., Van Roy, V., Vertesy, D., & Vivarelli, M. (2025). Is artificial intelligence leading to a new
10 technological paradigm? *Structural Change and Economic Dynamics*, 72, 347–359.
11 <https://doi.org/10.1016/j.strueco.2024.12.006>
- 12 *Documentation and downloads | Human Development Reports*. (n.d.). Retrieved February 11, 2026,
13 from <https://hdr.undp.org/data-center/documentation-and-downloads>
- 14 *Endogenous Technological Change by Paul M. Romer: SSRN*. (n.d.). Retrieved May 14, 2026, from
15 https://papers.ssrn.com/sol3/papers.cfm?abstract_id=226703
- 16 *Fixed Effects Estimation of Large-TPanel Data Models | Annual Reviews*. (n.d.). Retrieved May 15, 2026,
17 from [https://www.annualreviews.org/content/journals/10.1146/annurev-economics-080217-](https://www.annualreviews.org/content/journals/10.1146/annurev-economics-080217-053542)
18 [053542](https://www.annualreviews.org/content/journals/10.1146/annurev-economics-080217-053542)
- 19 Freire, C. (2025). Is this time different? Impact of AI in output, employment and inequality across low,
20 middle and high-income countries. *Structural Change and Economic Dynamics*, 73, 136–157.
21 <https://doi.org/10.1016/j.strueco.2024.12.016>

- 1 GDP per capita growth (annual %) | Data. (n.d.). Retrieved February 11, 2026, from
2 [https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG?end=2024&name_desc=false&start=](https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG?end=2024&name_desc=false&start=1961&view=chart)
3 [1961&view=chart](https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG?end=2024&name_desc=false&start=1961&view=chart)
- 4 Goel, R. K., & Nelson, M. A. (2025). Awareness of artificial intelligence: Diffusion of AI versus ChatGPT
5 information with implications for entrepreneurship. *The Journal of Technology Transfer*, 50(1),
6 96–113. <https://doi.org/10.1007/s10961-024-10089-3>
- 7 Golic, Z. (2019). Finance and Artificial Intelligence: The Fifth Industrial Revolution and Its Impact on the
8 Financial Sector. *Proceedings of the Faculty of Economics in East Sarajevo / Zbornik Radova*
9 *Ekonomskog Fakulteta u Istočnom Sarajevu*, 19, 67–81. (2191760).
- 10 Gomes, O. (2025). The Human Capital—Artificial Intelligence Symbiosis and Economic Growth. *De*
11 *Economist*, 173(2), 331–365. <https://doi.org/10.1007/s10645-025-09452-y>
- 12 Khan, M. S., Umer, H., & Faruqe, F. (2024). Artificial intelligence for low income countries. *Humanities*
13 *and Social Sciences Communications*, 11(1), 1422. <https://doi.org/10.1057/s41599-024-03947-w>
- 14 Liu, F., & Liang, C. (2025). Analyzing wealth distribution effects of artificial intelligence: A dynamic
15 stochastic general equilibrium approach. *Heliyon*, 11(2), e41943.
16 <https://doi.org/10.1016/j.heliyon.2025.e41943>
- 17 MIAH, M. (2025). Ethical AI for Inclusive Economic Growth. *Informatica Economica*, 29(4), 35–46.
18 (190974277). <https://doi.org/10.24818/issn14531305/29.4.2025.04>
- 19 Minniti, A., Prettnner, K., & Venturini, F. (2025). AI innovation and the labor share in European regions.
20 *European Economic Review*, 177, 105043. <https://doi.org/10.1016/j.euroecorev.2025.105043>
- 21 Rogers, E. M. (1983). *Diffusion of innovations* (3. ed). Free Press [u.a.].

- 1 Singh, S. (2025, November 20). ChatGPT Users Statistics (February 2026) – Growth & Usage Data.
- 2 DemandSage. <https://www.demandsage.com/chatgpt-statistics/>
- 3 *The Impact of Artificial Intelligence on Innovation* | NBER. (n.d.). Retrieved May 22, 2026, from
- 4 <https://www.nber.org/papers/w24449>
- 5 *The Productivity J-Curve: How Intangibles Complement General Purpose Technologies—American*
- 6 *Economic Association*. (n.d.). Retrieved May 22, 2026, from
- 7 <https://www.aeaweb.org/articles?id=10.1257/mac.20180386>
- 8 TURING, A. M. (1950). I.—COMPUTING MACHINERY AND INTELLIGENCE. *Mind*, LIX(236), 433–460.
- 9 <https://doi.org/10.1093/mind/LIX.236.433>
- 10 *Who on Earth Is Using Generative AI ?* (2024). <https://doi.org/10.1596/1813-9450-10870>
- 11 *World Development Indicators* | DataBank. (n.d.). Retrieved February 24, 2026, from
- 12 <https://databank.worldbank.org/source/world-development-indicators#>
- 13 Yang, C.-H. (2022). How Artificial Intelligence Technology Affects Productivity and Employment: Firm-
- 14 level Evidence from Taiwan. *Research Policy*, 51(6), 104536.
- 15 <https://doi.org/10.1016/j.respol.2022.104536>
- 16 For all calculations and coding here is the publicly available GitHub link:
- 17 <https://github.com/MatthiasRcode/Matyas-Thesis.git>

1

Planning

Time Periods	Key goals	Deadlines
Febr 23 - March 4	Polishing the draft of the proposal according to the comments that I have received and having my first regression results (just to see the possibilities).	March 4 - Research Proposal
March 5 - Apr 17	Working on the thesis again based on the feedback from the supervisor. Improving the regression model and results.	Apr 17 - First Draft
Apr 18 - May 20	Revisiting all sections of the paper and changing it according to the preliminary results.	
May 21 - 29	Finishing the draft version of the paper, this week it should only be some minor changes.	May 29 – Thesis Draft
May 30 - Jun 3	Making the presentation based on the paper.	Jun 3 - Presentation
Jun 4 - 12	Correcting the paper, if necessary, based on feedback.	Jun 12 - Final Thesis

2